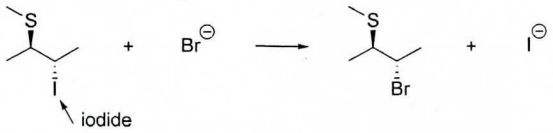
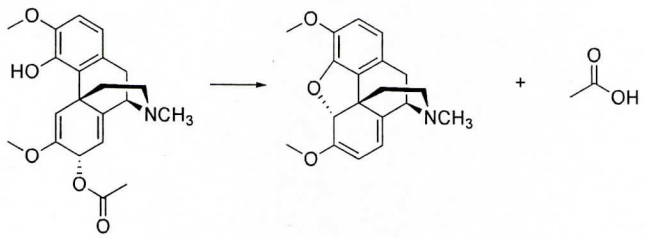
Organic Cume 03-02-2018

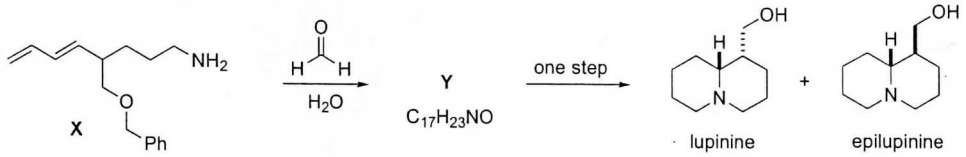
1. Propose a mechanism to account for the following transformation including the stereochemistry.



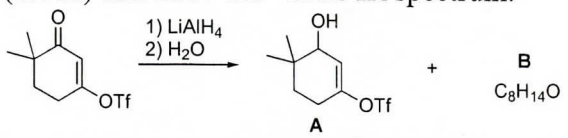
2. The following is a reaction in the biosynthesis of morphine in opium poppies. Draw a complete mechanism that should account for the stereochemistry.



3. Heating compound **X** with aqueous formaldehyde forms **Y** ($C_{17}H_{23}NO$), which has been converted to a mixture of lupinine and epilupinine, alkaloids isolated from lupin, a perennial ornamental plant commonly seen on the roadside in parts of Alaska. Identify **Y** and explain how it is formed.



4. Treatment of the following ketone with $LiAlH_4$ affords two products, **A** and **B**. Compound **B** has the molecular formula $C_8H_{14}O$ and exhibits strong signals at 3305 cm^{-1} (broad) and 2117 cm^{-1} in its IR spectrum:



(a) Using the following ^1H NMR data, deduce the structure of compound **B**: 0.89 δ (6H, singlet), 1.49 δ (1H, broad singlet), 1.56 δ (2H, triplet), 1.95 δ (1H, singlet), 2.19 δ (2H, triplet), 3.35 δ (2H, singlet).

(b) Provide a plausible mechanism to account for the formation of compound **B** (Tf = triflate).